

08.1	1. (water has a relatively) high (specific) heat capacity; 2. Can gain / lose a lot of heat / energy without changing temperature; OR Takes a lot of heat / energy to change temperature;	2	1. Ignore numbers relating to heat capacity 2. Accept due to H bonding between water molecules
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08.2	Adenosine diphosphate and (inorganic) phosphate;	1	Accept ADP for adenosine diphosphate Accept Pi / PO_4^{3-} / P in a circle for inorganic phosphate Reject adenine diphosphate Reject phosphorus / P for phosphate
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08.3	1. Species / organism the muscle tissue came from; OR Thickness / type / source of the muscle tissue; 2. Temperature of the muscle tissue / <u>ATP</u> solution / slides; 3. pH of the <u>ATP</u> solution;	2 max	1. Ignore surface area of muscle tissue 2. and 3. Need to be qualified Reject concentration / volume of ATP hydrolase
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08.4	Description 1. As concentration of ATP increases, length of muscle decreases; Explanation 2. More ATP (hydrolysed by ATP hydrolase), so more energy released, so more muscle contraction / shortening of muscle;	2	1. Accept negative correlation 2. Accept more ATP available for correct/named aspect of muscle contraction 2. Idea of more is required once. 2. Reject energy produced
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08.5	4.88×10^{-6} ;;; If answer incorrect EITHER Allow 1 mark for 0.244 Allow 1 mark for 1.22×10^{-5} OR Allow 1mark for 12200 / 1.525 Allow 1 mark for 0.61	3	Accept 5×10^{-6} Accept correct answer however expressed Max 2 for incorrect final answer
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